Clustering is also used as a preliminary step for other more sophisticated algorithms, because it provides a cheap way to break down large datasets into coherent groups.

But we'll see this and lot more in next chapter.

11.5 Summary

- Nearest neighbor search can be used to improve geographical matching of physical resources: for example, finding the closest shop to a customer.
- When we move from theory to real-world applications, we have to take many factors into account and adjust our NN search algorithm to take into account business logic, for instance allowing filtering of the resources among which we search or weighting the results according to some business rules.
- We also have to deal with the limitations of physical IT systems, including memory restrictions, CPU availability, and network constraints. Often this means we will have to compromise about algorithm's precision in order to achieve acceptable response times.
- Distributed web applications impose new issues that needs to be taken into account when designing a system: it's not enough to come up with a good algorithm, we need to choose/design one that also works in the real system we are building.
- NN search is useful in a number of other areas, from simulations in particle physics to machine learning.